

Last name _____

First name _____

LARSON—MATH 756—CLASSROOM WORKSHEET 13

Lovász Theta

Review

1. Why does $\alpha = \alpha_f$ for bipartite graphs?
2. What graph classes generalize KE graphs? Ideally we'd like a larger class where (1) it is possible to efficiently recognize that a graph is in the class, and (2) where α can be computed efficiently.
3. What is an *orthonormal representation* of a graph?
4. Find an orthonormal representation for K_3 .
5. Find an orthonormal representation for C_4 .
6. Find $\vartheta(C_4)$.
7. Why is ϑ well-defined?

New

1. What is the definition of ϑ ?
2. Find $\vartheta(K_n)$.

3. Show: for any graph, $\alpha \leq \vartheta$.

4. Find $\vartheta(K_{2,3})$.

5. Find $\vartheta(K_{m,n})$ ($n \geq m$).

6. Find $\vartheta(G)$ for any bipartite graph $G = (A, B)$.